PAPER_673: UQFF THz Holes and Red Dwarf Reactor Meta-Module

Subtitle: Synthesises Session 172 advancements: THz superconductor BH analogy, Red Dwarf UQFF reactor, Framework Advancement Score, and self-consistent cycle. **Module:** UQFFAdvancementsAndTHzHoles
Session: Session 172
Date: April 2, 2026
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Status: Complete — CP4 #257 | UQFF Session 172

Abstract

This meta-module synthesises the major UQFF framework advancements from Session 172 (PAPER_657–673). We introduce the THz Hole analogy, the Red Dwarf Reactor UQFF model, the Framework Advancement Score (FAS), and document the self-consistent UQFF cycle linking all 17 new papers.

1. THz Hole Analogy

Superconductors at T_c ~ 100 K exhibit quasi-particle "holes" traversing the condensate, analogous to the Hawking pair mechanism.

$$f_{THz} = \frac{k_B T_c}{2\pi\hbar} \approx 2\,\text{THz at } T_c = 100\,\text{K}$$

UQFF maps: ρ_SCm / (m_e c^2) $\leftrightarrow$ Cooper pair density at horizon analogue.

$$L_{THz,UQFF} = L_H \cdot \left(\frac{f_{THz}}{f_{Hawking}}\right)^4 \cdot \frac{\rho_{UA}}{\rho_{SCm}}$$

2. Red Dwarf Reactor UQFF

Red dwarf core: T_core ~ 10^7 K, ρ_core ~ 10^5 kg/m^3.

$$\Gamma_{UQFF} = \sigma_{pp} n^2 \left(1 - \frac{\rho_{SCm}}{\rho_{UA}}\right)$$

Lifetime extension:

$$\tau_{RD,UQFF} = \tau_{RD,std} \cdot \frac{\rho_{UA}}{\rho_{SCm}} \cdot (1 + f_{TRZ}) \approx 1.1 \times 10^{14}\,\text{yr}$$

For comparison, τ_std ~ 10^{13} yr.

3. Framework Advancement Score (FAS)

$$FAS = N_{papers} \cdot (1 + f_{TRZ}) \cdot \sqrt{\frac{\rho_{UA}}{\rho_{SCm}}}$$

At N = 673: **FAS ≈ 2341**

Tracks the pace of UQFF physics discovery normalised to vacuum coupling constants.

4. Self-Consistent Cycle (PAPER_657–673)

```
KB7(657) → BH_Bounce(658) → BH_WH(659) → WH_Rad(660)
→ PBH_DM(661) → Hawking(662) → BH_Inv(663) → WH_Stab(664)
→ Suppress(665) → GW_Supp(666) → BH_Stab(667) → PBH_Stab(668)
→ GW150914(669) → Accretion(670) → dM/dt(671) → tau_evap(672)
→ THz_Holes(673) → back to KB7
```

5. C++ Module

`UQFFAdvancementsAndTHzHoles.h / .cpp` — Session 172 CP4 #257 — `UQFFAdvancementsAndTHzHolesCalculator`

Session 172 Summary

Metric	Value
Papers created	PAPER_660–673 (14 new)
Total papers	673 / 1000 (67.3%)
CP4 entries	257
C++ module pairs	+14 (80 total)
Session version	v5.29

PAPER_673 | Session 172 | Star-Magic UQFF Framework v5.29 | Daniel Murphy